

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12405455
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Sakamoto; Masaru

Zoom lens and image pickup apparatus

Abstract

A zoom lens includes, from an object side to an image side, a positive first lens unit configured not to move for zooming, at least three movable lens units configured to move for zooming, and a positive final lens unit closest to the image side in lens units and configured not to move for zooming. In the at least three movable lens units, a lens unit closest to the image side has a positive power, and a lens unit second closest to the image side has a negative power. The first lens unit includes a first sub unit closest to the object side and configured not to move for focusing and a second sub unit configured to move for focusing. The final lens unit includes a positive lens closest to the image side in elements having a refractive power in the final lens unit. Predetermined inequalities are satisfied.

Inventors:	Sakamoto; Masaru (Tochigi, JP)
Applicant:	CANON KABUSHIKI KAISHA (Tokyo, JP)
Family ID:	82898749
Assignee:	Canon Kabushiki Kaisha (Tokyo, JP)
Appl. No.:	17/818525
Filed:	August 09, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230064725 A1	Mar. 02, 2023

Foreign Application Priority Data

JP	2021-131582	Aug. 12, 2021
----	-------------	---------------

Publication Classification

Int. Cl.: G02B15/14 (20060101); G02B15/167 (20060101); G02B15/20 (20060101)

U.S. Cl.:

CPC G02B15/145125 (20190801); G02B15/1461 (20190801); G02B15/167 (20130101); G02B15/20 (20130101);

Field of Classification Search

CPC: G02B (15/145125); G02B (15/145127); G02B (15/1461); G02B (15/167); G02B (15/20)

USPC: 359/683; 359/684

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2014/0204252	12/2013	Yoshimi	359/683	G02B 15/1461
2015/0301319	12/2014	Komatsu	359/684	G02B 27/1013
2016/0062091	12/2015	Wada	359/683	G02B 15/145121
2016/0062096	12/2015	Hatada	348/240.3	G02B 27/0025
2016/0091698	12/2015	Ikeda	359/683	G02B 15/1461
2018/0143412	12/2017	Sakamoto	N/A	G02B 15/17
2021/0048654	12/2020	Ogawa	N/A	G02B 15/145125
2021/0048655	12/2020	Shimomura	N/A	N/A
2021/0278641	12/2020	Taki	N/A	G02B 15/144511

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
------------	------------------	---------	-----

2016-012118	12/2015	JP	N/A
2021021902	12/2020	JP	N/A
2021-032925	12/2020	JP	N/A

OTHER PUBLICATIONS

Warren J Smith; "Modern Lens Design;" Genesse Optics Software, Inc. Rochester, New York; Jan. 1, 1992; pp. 1-5. cited by applicant
Herbert Gross; "Handbook of Optical Systems;" vol. 3: Aberration Theory and Correction of Optical Systems; Jan. 1, 2007; pp. 1-6. cited by applicant

Primary Examiner: Won; Bumsuk

Assistant Examiner: Moser; Seth D

Attorney, Agent or Firm: Canon U.S.A., Inc. IP Division

Background/Summary

BACKGROUND

Technical Field

(1) The aspect of the embodiments relates to a zoom lens and an image pickup apparatus.

Description of the Related Art

(2) Zoom lenses are in demand to have high resolution uniformly from center to periphery of their angle of views and to cause less chromatic aberration. Japanese Patent Laid-Open No. 2016-012118 discloses a positive lead type zoom lens including, in order from an object side to an image side, a positive lens unit that does not move during zooming, a negative lens unit that moves during zooming, and a positive lens unit that moves during zooming.

(3) Zoom lenses are also in demand to have wide angles, wide magnification ranges (high zoom ratios), high aperture ratios, and small sizes. Positive lead type zoom lenses are beneficial for increasing the zoom ratio as compared with negative lead type zoom lenses in each of which a first lens unit has a negative refractive index. However, in order that a positive lead type zoom lens is made to provide high optical performance, the positive lead type zoom lens is made to perform balanced correction on on-axis chromatic aberration and lateral chromatic aberration.

(4) If the number of lenses is reduced so that a size of a zoom lens is reduced, and a refractive power of each lens unit or of a single lens is increased, chromatic aberration and other various errors may increase. If a wider angle of view and a higher zoom ratio are to be acquired, it may be difficult for the zoom lens to correct various aberrations, especially lateral chromatic aberration at a wide-angle end.

SUMMARY

(5) A zoom lens according to one aspect of the embodiments includes, in order from an object side to an image side, a first lens unit having a positive refractive power and configured not to move for zooming, at least three movable lens units configured to move so that a distance between each pair of adjacent lens units therein changes for zooming, and a final lens unit disposed closest to the image side in lens units included in the zoom lens, having a positive refractive power, and configured not to move for zooming. A movable lens unit closest to the image side in the at least three movable lens units has a positive refractive power, and a movable lens unit second closest to the image side in the at least three movable lens units has a negative refractive power. The first lens unit includes a first sub unit closest to the object side and configured not to move for focusing, and a second sub unit configured to move for focusing. The final lens unit includes a positive lens closest to the image side in elements having a refractive power in the final lens unit. Following inequalities are satisfied:

$$62 \leq vdp \leq 81.54,$$

$$2.24 \leq Ndp + 0.01vdp, \text{ and}$$

$$4 \leq Z \leq 50$$

where vdp represents an Abbe number with respect to a d-line of the positive lens, Ndp represents a refractive index with respect to the d-line of the positive lens, and Z represents a zoom ratio of the zoom lens.

(6) An apparatus including the above zoom lens also constitute another aspect of the embodiments of the present disclosure.

(7) Further features of the disclosure will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a sectional view of a zoom lens according to Example 1 (Numerical Example 1).
- (2) FIGS. 2A and 2B are aberration diagrams at a wide-angle end and a telephoto end of the zoom lens according to Example 1.
- (3) FIG. 3 is a sectional view of a zoom lens according to Example 2 (Numerical Example 2).
- (4) FIGS. 4A and 4B are aberration diagrams at a wide-angle end and a telephoto end of the zoom lens according to Example 2.
- (5) FIG. 5 is a sectional view of a zoom lens according to Example 3 (Numerical Example 3).
- (6) FIGS. 6A and 6B are aberration diagrams at a wide-angle end and a telephoto end of the zoom lens according to Example 3.
- (7) FIG. 7 is a sectional view of a zoom lens according to Example 4 (Numerical Example 4).
- (8) FIGS. 8A and 8B are aberration diagrams at a wide-angle end and a telephoto end of the zoom lens according to Example 4.
- (9) FIG. 9 is a sectional view of a zoom lens according to Example 5 (Numerical Example 5).
- (10) FIGS. 10A and 10B are aberration diagrams at a wide-angle end and a telephoto end of the zoom lens according to Example 5.
- (11) FIG. 11 is a sectional view of a zoom lens according to Example 6 (Numerical Example 6).
- (12) FIGS. 12A and 12B are aberration diagrams at a wide-angle end and a telephoto end of the zoom lens according to Example 6.
- (13) FIG. 13 is a sectional view of a zoom lens according to Example 7 (Numerical Example 7).
- (14) FIGS. 14A and 14B are aberration diagrams at a wide-angle end and a telephoto end of the zoom lens according to Example 7.
- (15) FIG. 15 is a diagram illustrating an image pickup apparatus having the zoom lens according to any one of Examples 1 to 7.

DESCRIPTION OF THE EMBODIMENTS

(16) Referring now to the accompanying drawings, a description will be given of embodiments according to the present disclosure.

(17) First, prior to a specific description of Examples 1 to 7, a description is given of matters common to each example. A zoom lens according to each example is suitable as an image pickup optical system for an image pickup apparatus such as a still camera, a video camera, a TV camera, a cinema camera, and a monitoring camera. In the zoom lens according to each example, a lens unit or a sub unit is a group of one or a plurality of

lenses. The lenses integrally move as one while during zooming between a wide-angle end and a telephoto end and during focusing to an infinite distance side or to a close side. That is, a distance between each pair of adjacent lens units or sub units changes during zooming and focusing. A lens unit may include an aperture diaphragm. The wide-angle end represents a zooming state with a maximum angle of view (shortest focal length), and the telephoto end represents a zooming state with a minimum angle of view (maximum focal length). Each lens unit that moves during zooming is located, in the state of the wide-angle end, at one end and, in the state of the telephoto end, at the other end of a mechanical or control movable range on an optical axis.

(18) Next, a description is given of the zoom lens illustrated in FIG. 1 according to Example 1 as an example (with reference signs in parentheses).

(19) The zoom lens according to each example includes, in order from an object side to an image side, a first lens unit (U1) that has a positive refractive power and does not move (is fixed) during (or for) zooming, at least three movable lens units (U2 to U4) that move so that a distance between each pair of adjacent lens units therein changes during (or for) zooming, and a final lens unit (U5) that is closest to the image side, has a positive refractive power, and does not move during (or for) zooming. In the above-described at least three movable lens units (U2 to U4), a movable lens unit (U4) closest to the image side has a positive refractive power, and a movable lens unit (U3) second closest to the image side has a negative refractive power. The first lens unit (U1) includes a fixed sub unit (U11) that is closest to the object side and does not move during (or for) focusing, and a movable sub unit (U12 and U14) that moves during (or for) focusing.

(20) In this way, the zoom lens according to each example performs zooming by moving the at least three movable lens units (U2 to U4), increases a zoom ratio by using the negative movable lens unit (U3) second closest to the image side in the at least three movable lens units (U2 to U4), and effectively reduces (or corrects) an aberration variation during zooming by using the positive movable lens unit (U4) closest to the image side in the at least three movable lens units (U2 to U4). This makes it possible to realize a small-sized zoom lens that has a wide magnification range (high zoom ratio) and high optical performance.

(21) In addition, by moving a part of the first lens unit (U1) during focusing and disposing the fixed sub unit (U11) at a position closest to the object side in the first lens unit (U1), it is possible to reduce an aberration variation caused by focusing. Further, it is possible to reduce sizes of the movable sub unit (U12, U14) that moves during focusing and of its driving mechanism, and thereby both high zoom ratio and size reduction can be realized.

(22) The zoom lens according to each example further satisfies the following inequalities (1) to (3) in addition to the above-described configurations of the movable lens units and the first lens unit, so as to realize a zoom lens suitable for improving a definition and a function of an image pickup apparatus.

(23) In the final lens unit (U5), a lens closest to the image side in elements having refractive powers is a positive lens (U5Lp). vdp and Ndp respectively represent an Abbe number with respect to a d-line and a refractive index with respect to the d-line of the positive lens U5Lp, and Z represents the zoom ratio of the zoom lens. Here, the zoom lens according to each example satisfies the following inequalities (1) to (3).

$$62 \leq vdp \quad (1)$$

$$2.24 \leq Ndp + 0.01vdp \quad (2)$$

$$4 \leq Z \leq 5 \quad (3)$$

(24) The inequalities (1) and (2) represents conditions relating to characteristics of optical glass of the positive lens (U5Lp) closest to the image side in the final lens unit U5. Usually, optical glass contains a plurality of types of metal oxides such as SiO.sub.2, TiO.sub.2, La.sub.2O.sub.3, Al.sub.2O.sub.3, Nb.sub.2O.sub.5, ZrO.sub.2, and Gd.sub.2O.sub.3. Among them, TiO.sub.2 has an effect of increasing a refractive index and reducing an Abbe number. Therefore, optical glass with a high content of TiO.sub.2 has a relatively high refractive index and a high dispersion. Gd.sub.2O.sub.3 has an effect of increasing a refractive index and increasing an Abbe number. Optical glass with a high content of Gd.sub.2O.sub.3 has a relatively high refractive index and a low dispersion. TiO.sub.2 originally has a high refractive index and a high dispersion and Gd.sub.2O.sub.3 originally has a high refractive index and a low dispersion, and therefore the characteristics of optical glass containing them are similar to the original characteristics of the metal oxides.

(25) As described above, optical glass has a property such that its characteristics change depending on contents of substances (components), and thus desired optical characteristics (refractive index and Abbe number) can be acquired by properly setting amounts of the substances. The same similarly applies to optical ceramics. For example, if optical ceramics includes a high content of a substance having a high refractive index and a low dispersion, the optical ceramics have relatively high refractive index and low dispersion. Therefore, it is possible to acquire an optical material such as optical glass and optical ceramics having desired optical characteristics if amounts of substances and metal oxides such as SiO.sub.2, TiO.sub.2, and La.sub.2O.sub.3 are properly set and are dissolved or sintered.

(26) In the zoom lens having the above configurations, as the angle of view becomes wider, a refractive power of a negative lens (U1Ln) closest to the object side in the first lens unit U1 becomes stronger. Further, at the wide-angle end, a height increases of an off-axis principal ray entering the negative lens (U1Ln) closest to the object side in the first lens unit U1, and as a result, a generated amount of lateral chromatic aberration increases. Thus, the wider angle of view makes it difficult to correct the lateral chromatic aberration especially at the wide-angle end.

(27) Further, as the zoom ratio becomes higher, a height of an on-axis ray in the first lens unit (U1) increases in proportion to a focal length, and as a result, on-axis chromatic aberration occurring in the first lens unit (U1) increases. In order that the on-axis chromatic aberration occurring in the first lens unit (U1) is reduced, optical glass having a high anomalous dispersion in a movable lens unit that has a strong negative refractive power and moves during zooming is used. A height of an off-axis principal ray entering the movable lens unit at the wide-angle end is high, and therefore large lateral chromatic aberration is caused by the optical glass having the high anomalous dispersion in the movable lens unit. Thus, increasing the zoom ratio also makes it difficult to correct the lateral chromatic aberration at the wide-angle end.

(28) In general, a lens placed at a position at which a height of an off-axis ray is high has a large contribution to correction of lateral chromatic aberration. For correcting well the lateral chromatic aberration at the wide-angle end that occurs with widening the angle of view and increasing the zoom ratio, it is effective to use an optical material having a low dispersion and a high anomalous dispersion in the positive lens (U5Lp) at a position farthest from the aperture. In this positive lens (U5Lp), a chief paraxial ray, which has a large effect on lateral chromatic aberration, passes through a position where a height from the optical axis is high. Therefore, using the above-described optical material for the positive lens (U5Lp) makes larger the effect on lateral chromatic aberration, and thus the lateral chromatic aberration can be corrected well.

(29) The inequality (1) represents a condition relating to the Abbe number of the positive lens (U5Lp) closest to the image side in the final lens unit (U5). If vdp is smaller than the lower limit of the inequality (1), it is difficult to correct lateral chromatic aberration that occurs in the first lens unit (U1) and is insufficiently corrected by the first lens unit (U1). In addition, such an optical glass material makes the positive lens (U5Lp) have a small partial dispersion ratio θ_gF , which makes it difficult to correct a secondary spectrum of lateral chromatic aberration occurring in a movable lens unit having a strong negative refractive power that moves during zooming.

(30) The numerical range of the inequality (1) may be set to a numerical range of the following inequality (4).

$$80 \leq vdp \quad (4)$$

(31) The numerical range of the inequality (4) may be set to the following numerical range.

$$80 \leq vdp \leq 97 \quad (4a)$$

(32) If vdp is larger than the upper limit of the inequality (4a), it is difficult to produce a low-dispersion optical glass material.

(33) The inequality (2) represents a condition relating to a relationship between the Abbe number and the refractive index of the positive lens (U5Lp). If $Ndp + 0.01 vdp$ is smaller than the lower limit of the inequality (2), such an optical glass material makes the positive lens (U5Lp) have a small partial dispersion ratio θ_gF , which makes it difficult to correct a secondary spectrum of lateral chromatic aberration occurring in a movable lens unit having a strong negative refractive power that moves during zooming.

(34) The numerical range of the inequality (2) may be set to the following numerical range.

$$2.24 \leq Ndp + 0.01vdp \leq 2.50 \quad (2a)$$

(35) If $Ndp + 0.01vdp$ is larger than the upper limit of the inequality (2a), it is difficult to produce an optical glass material having a low dispersion and a high refractive index.

(36) The numerical range of the inequality (2) may be set to the following numerical range.

$$2.24 \leq Ndp + 0.01vdp \leq 2.40 \quad (2b)$$

(37) The inequality (3) represents a condition for realizing a zoom lens having a large zoom ratio while having a small size. If Z is smaller than the lower limit of the inequality (3), it is difficult to increase the zoom ratio of the zoom lens. If Z is larger than the upper limit of the inequality (3), it is difficult to reduce the size of the zoom lens.

(38) The numerical range of the inequality (3) may be set to the following numerical range.

$$4.3 \leq Z \leq 47.0 \quad (3a)$$

(39) The numerical range of the inequality (3) may be set to the following numerical range.

$$4.4 \leq Z \leq 43.0 \quad (3b)$$

(40) If the above-described conditions (inequalities) are satisfied, it is possible to realize a small-sized and light-weighted zoom lens having a wide angle of view, a large diameter, a high zoom ratio, and good optical performance.

(41) The zoom lens according to each example may satisfy at least one of the following conditions (inequalities). In the zoom lens according to each example, the final lens unit (U5) may include four or more lenses each of which has a refractive power. When the number of lenses included in the final lens unit (U5) is properly set, it is possible to correct well various aberrations at the wide-angle end. The final lens unit (U5) may include at least six or more lenses each of which has a refractive power.

(42) The zoom lens according to each example may satisfy the following inequality (5) where f_p represents a focal length of the positive lens (U5Lp) closest to the image side in the final lens unit (U5), and f_v represents a focal length of the lens unit (U2) having the strongest negative refractive power in the at least three movable lens units (U2 to U4).

$$0.5 \leq |f_p/f_v| \leq 50.0 \quad (5)$$

(43) If $|f_p/f_v|$ is larger than the upper limit of the inequality (5), the refractive power of the positive lens (U5Lp) is so weak that it becomes difficult to correct the lateral chromatic aberration at the wide-angle end. If $|f_p/f_v|$ is smaller than the lower limit of the inequality (5), the refractive power of the positive lens (U5Lp) is so strong that the lateral chromatic aberration at the wide-angle end is excessively corrected. Further, since the refractive power of the positive lens (U5Lp) is strong, various aberrations occurring in the final lens unit (U5) increase, which makes it difficult to correct various aberrations in the wide-angle end.

(44) The numerical range of the inequality (5) may be set to the following numerical range.

$$0.7 \leq |f_p/f_v| \leq 40.0 \quad (5a)$$

(45) The numerical range of the inequality (5) may be set to the following numerical range.

$$0.85 \leq |f_p/f_v| \leq 35.00 \quad (5b)$$

(46) The zoom lens according to each example may satisfy the following inequality (6) where L_e represents a distance on the optical axis from a lens surface closest to the object side in the final lens unit (U5) to a lens surface closest to the image side in the final lens unit (U5), and f_w represents a focal length at the wide-angle end of the zoom lens.

$$2.5 \leq L_e/f_w \leq 25.0 \quad (6)$$

(47) If L_e/f_w is larger than the upper limit of the inequality (6), a height of an off-axis ray at the positive lens (U5Lp) is so high that it becomes difficult to reduce the size of the positive lens (U5Lp). If L_e/f_w is smaller than the lower limit of the inequality (6), the height of the off-axis ray at the positive lens (U5Lp) is so low that it becomes difficult to correct the lateral chromatic aberration at the wide-angle end.

(48) The numerical range of the inequality (6) may be set to the following numerical range.

$$3.0 \leq L_e/f_w \leq 20.0 \quad (6a)$$

(49) The numerical range of the inequality (6) may be set to the following numerical range.

$$3.5 \leq L_e/f_w \leq 17.0 \quad (6b)$$

(50) In the zoom lens according to each example, the movable lens unit (U2) having the strongest negative refractive power in the at least three movable lens units (U2 to U4) each of which moves during zooming may include at least two negative lenses and one positive lens. Thereby, it is possible to realize a small zoom lens that has a high zoom ratio and high optical performance and is suitable for improving a definition and a function of an image pickup apparatus.

(51) In the zoom lens according to each example, the first lens unit (U1) may include a positive lens (U1Lp) at a position closest to the image side. This makes it possible to realize a small zoom lens having a wide angle of view.

(52) In the zoom lens according to each example, the lens unit (U2) having the strongest negative refractive power in the at least three movable lens units each of which moves during zooming may move to the image side during zooming from the wide-angle end to the telephoto end. This makes it possible to realize a small zoom lens having a high zoom ratio and high optical performance.

(53) Hereinafter, a specific description is given of Examples 1 to 7. After the description of Example 7, Numerical Examples 1 to 7 corresponding to Examples 1 to 7 are given.

Example 1

(54) A zoom lens according to Example 1 (Numerical Example 1) illustrated in FIG. 1 includes, in order from an object side to an image side, a first lens unit U1 having a positive refractive power, a second lens unit U2 having a negative refractive power, a third lens unit U3 having a negative refractive power, a fourth lens unit U4 having a positive refractive power, and a fifth lens unit U5 having a positive refractive power. FIG. 1 illustrates a lens structure at a wide-angle end and at a state of focusing on an object at an infinite distance. The zoom lens according to Numerical Example 1 has a zoom ratio of 9.8 and satisfies the inequality (3).

(55) The first lens unit U1 is a lens unit fixed (that does not move) during zooming. The second to fourth lens units U2 to U4 are movable lens units each of which moves during zooming. As indicated by arrows below the second to fourth lens units U2 to U4 in the drawing, during zooming from the wide-angle end to the telephoto end, the second lens unit U2 moves to the image side, and the third lens unit U3 and fourth lens unit U4 move to the object side and then move to the image side. The fourth lens unit U4 includes an aperture diaphragm SP at a position close to the object side. The fifth lens unit U5 is a final lens unit fixed during zooming. IP represents an image plane. An image plane of an image sensor (image pickup element or photoelectric conversion element) or a photosensitive surface of a silver halide film is disposed on the image plane IP.

(56) In the first lens unit U1, a sub unit U11 closest to the object side and a sub unit U13 third closest to the object side are fixed sub units each of which is fixed during focusing, and a sub unit U12 second closest to the object side and a sub unit U14 closest to the image side are movable sub units each of which moves during focusing. As indicated by arrows below the movable sub units U12 and U14 in the drawing, during focusing from an infinite distance side to a close side, the movable sub unit U12 moves to the image side, and the movable sub unit U14 moves to the object side.

(57) In Numerical Example 1, the positive lens U5Lp closest to the image side in the fifth lens unit U5 satisfies the inequalities (1) to (2), the inequality (4), and the inequality (6). The movable lens unit having the strongest negative refractive power in the movable lens units is the second lens unit U2. The second lens unit U2 satisfies the inequality (5).

(58) The first lens unit U1 includes a negative lens U1Ln at a position closest to the object side and a positive lens U1Lp at a position closest to the image side.

Example 2

(59) A zoom lens according to Example 2 (Numerical Example 2) illustrated in FIG. 3 includes, in order from an object side to an image side, a first lens unit U1 having a positive refractive power, a second lens unit U2 having a negative refractive power, a third lens unit U3 having a negative refractive power, a fourth lens unit U4 having a positive refractive power, and a fifth lens unit U5 having a positive refractive power. FIG. 3 illustrates a lens structure at a wide-angle end and at a state of focusing on an object at an infinite distance. The zoom lens according to Numerical Example 2 has a zoom ratio of 10.0 and satisfies the inequality (3). P in the drawing represents an optical element having no refractive power such as a prism.

(60) The first lens unit U1 is a lens unit fixed during zooming. The second to fourth lens units U2 to U4 are movable lens units each of which moves during zooming. As indicated by arrows below the second to fourth lens units U2 to U4 in the drawing, during zooming from the wide-angle end to the telephoto end, the second lens unit U2 moves to the image side, and the third lens unit U3 and fourth lens unit U4 move to the object side and then move to the image side. The fourth lens unit U4 includes an aperture diaphragm SP at a position close to the object side. The fifth lens unit U5 is a final lens unit fixed during zooming.

(61) In the first lens unit U1, a sub unit U1f closest to the object side is a fixed sub unit that is fixed during focusing, and a sub unit U12 second closest to the object side (closest to the image side) is a movable sub unit that moves during focusing. As indicated by an arrow below the movable sub unit U12 in the drawing, the movable sub unit U12 moves to the object side during focusing from an infinite distance side to a close side.

(62) In Numerical Example 2, the positive lens U5Lp closest to the image side in the fifth lens unit U5 satisfies the inequalities (1) to (2), the inequality (4), and the inequality (6). The movable lens unit having the strongest negative refractive power in the movable lens units is the second lens unit U2. The second lens unit U2 satisfies the inequality (5).

(63) The first lens unit U1 includes a negative lens U1Ln at a position closest to the object side and a positive lens U1Lp at a position closest to the image side.

Example 3

(64) A zoom lens according to Example 3 (Numerical Example 3) illustrated in FIG. 5 includes, in order from an object side to an image side, a first lens unit U1 having a positive refractive power, a second lens unit U2 having a positive refractive power, a third lens unit U3 having a negative refractive power, a fourth lens unit U4 having a negative refractive power, a fifth lens unit U5 having a positive refractive power, and a sixth lens unit U6 having a positive refractive power. FIG. 5 illustrates a lens structure at a wide-angle end and at a state of focusing on an object at an infinite distance. The zoom lens according to Numerical Example 3 has a zoom ratio of 20.0 and satisfies the inequality (3).

(65) The first lens unit U1 is a lens unit fixed during zooming. The second to fifth lens units U2 to U5 are movable lens units each of which moves during zooming. As indicated by arrows below the second to fifth lens units U2 to U5 in the drawing, during zooming from the wide-angle end to the telephoto end, the second lens unit U2 and the third lens unit U3 move to the image side, and the fourth lens unit U4 and the fifth lens unit U5 move to the object side and then move to the image side. The fifth lens unit U5 includes an aperture diaphragm SP at a position close to the image side. The sixth lens unit U6 is a final lens unit fixed during zooming.

(66) In the first lens unit U1, a sub unit U1f closest to the object side is a fixed sub unit that is fixed during focusing, and a sub unit U12 second closest to the object side (closest to the image side) is a movable sub unit that moves during focusing. As indicated by an arrow below the movable sub unit U12 in the drawing, the movable sub unit U12 moves to the object side during focusing from an infinite distance side to a close side.

(67) In Numerical Example 3, a positive lens U6Lp closest to the image side in the sixth lens unit U6 satisfies the inequalities (1) to (2), the inequality (4), and the inequality (6). The movable lens unit having the strongest negative refractive power in the movable lens units is the third lens unit U3. The third lens unit U3 satisfies the inequality (5).

(68) The first lens unit U1 includes a negative lens U1Ln at a position closest to the object side and a positive lens U1Lp at a position closest to the image side.

Example 4

(69) A zoom lens according to Example 4 (Numerical Example 4) illustrated in FIG. 7 includes, in order from an object side to an image side, a first lens unit U1 having a positive refractive power, a second lens unit U2 having a negative refractive power, a third lens unit U3 having a negative refractive power, a fourth lens unit U4 having a positive refractive power, and a fifth lens unit U5 having a positive refractive power. FIG. 7 illustrates a lens structure at a wide-angle end and at a state of focusing on an object at an infinite distance. The zoom lens according to Numerical Example 4 has a zoom ratio of 17.0 and satisfies the inequality (3).

(70) The first lens unit U1 is a lens unit fixed during zooming. The second to fourth lens units U2 to U4 are movable lens units each of which moves during zooming. As indicated by arrows below the second to fourth lens units U2 to U4 in the drawing, during zooming from the wide-angle end to the telephoto end, the second lens unit U2 moves to the image side, and the third lens unit U3 and fourth lens unit U4 move to the object side and then move to the image side. The fourth lens unit U4 includes an aperture diaphragm SP at a position close to the object side. The fifth lens unit U5 is a final lens unit fixed during zooming.

(71) In the first lens unit U1, a sub unit U1f closest to the object side is a fixed sub unit that is fixed during focusing, and a sub unit U12 second closest to the object side (closest to the image side) is a movable sub unit that moves during focusing. As indicated by an arrow below the movable sub unit U12 in the drawing, the movable sub unit U12 moves to the object side during focusing from an infinite distance side to a close side.

(72) In Numerical Example 4, the positive lens U5Lp closest to the image side in the fifth lens unit U5 satisfies the inequalities (1) to (2), the inequality (4), and the inequality (6). The movable lens unit having the strongest negative refractive power in the movable lens units is the second lens unit U2. The second lens unit U2 satisfies the inequality (5).

(73) The first lens unit U1 includes a negative lens U1Ln at a position closest to the object side and a positive lens U1Lp at a position closest to the image side.

Example 5

(74) A zoom lens according to Example 5 (Numerical Example 5) illustrated in FIG. 9 includes, in order from an object side to an image side, a first lens unit U1 having a positive refractive power, a second lens unit U2 having a negative refractive power, a third lens unit U3 having a negative refractive power, a fourth lens unit U4 having a positive refractive power, and a fifth lens unit U5 having a positive refractive power. FIG. 9 illustrates a lens structure at a wide-angle end and at a state of focusing on an object at an infinite distance. The zoom lens according to Numerical Example 5 has a zoom ratio of 4.5 and satisfies the inequality (3).

(75) The first lens unit U1 is a lens unit fixed during zooming. The second to fourth lens units U2 to U4 are movable lens units each of which moves during zooming. As indicated by arrows below the second to fourth lens units U2 to U4 in the drawing, during zooming from the wide-angle end to the telephoto end, the second lens unit U2 moves to the image side, and the third lens unit U3 and fourth lens unit U4 move to the object side and then move to the image side. The fourth lens unit U4 includes an aperture diaphragm SP at a position close to the object side. The fifth lens unit U5 is a final lens unit fixed during zooming.

(76) In the first lens unit U1, a sub unit U1f closest to the object side and a sub unit U13 third closest to the object side (closest to the image side) are fixed sub units each of which is fixed during focusing, and a sub unit U12 second closest to the object side is a movable sub unit that moves during focusing. As indicated by an arrow below the movable sub unit U12 in the drawing, the movable sub unit U12 moves to the image side during focusing from an infinite distance side to a close side.

(77) In Numerical Example 5, the positive lens U5Lp closest to the image side in the fifth lens unit U5 satisfies the inequalities (1) to (2), the inequality (4), and the inequality (6). The movable lens unit having the strongest negative refractive power in the movable lens units is the second

lens unit U2. The second lens unit U2 satisfies the inequality (5).

(78) The first lens unit U1 includes a negative lens U1Ln at a position closest to the object side and a positive lens U1Lp at a position closest to the image side.

Example 6

(79) A zoom lens according to Example 6 (Numerical Example 6) illustrated in FIG. 11 includes, in order from an object side to an image side, a first lens unit U1 having a positive refractive power, a second lens unit U2 having a negative refractive power, a third lens unit U3 having a negative refractive power, a fourth lens unit U4 having a negative refractive power, a fifth lens unit U5 having a positive refractive power, and a sixth lens unit U6 having a positive refractive power. FIG. 11 illustrates a lens structure at a wide-angle end and at a state of focusing on an object at an infinite distance. The zoom lens according to Numerical Example 6 has a zoom ratio of 40.0 and satisfies the inequality (3).

(80) The first lens unit U1 is a lens unit fixed during zooming. The second to fifth lens units U2 to U5 are movable lens units each of which moves during zooming. As indicated by arrows below the second to fifth lens units U2 to U5 in the drawing, during zooming from the wide-angle end to the telephoto end, the second lens unit U2 and the third lens unit U3 move to the image side and the fourth lens unit U4 and the fifth lens unit U5 move to the object side and then move to the image side. The sixth lens unit U6 is a final lens unit fixed during zooming and includes an aperture diaphragm SP at a position close to the object side.

(81) In the first lens unit U1, a sub unit U1f closest to the object side is a fixed sub unit that is fixed during focusing, and a sub unit U12 second closest to the object side (closest to the image side) is a movable sub unit that moves during focusing. As indicated by an arrow below the movable sub unit U12 in the drawing, the movable sub unit U12 moves to the object side during focusing from an infinite distance side to a close side.

(82) In Numerical Example 6, a positive lens U6Lp closest to the image side in the sixth lens unit U6 satisfies the inequalities (1) to (2), the inequality (4) and the inequality (6). The movable lens unit having the strongest negative refractive power in the movable lens units is the third lens unit U3. The third lens unit U3 satisfies the inequality (5).

(83) The first lens unit U1 includes a negative lens U1Ln at a position closest to the object side and a positive lens U1Lp at a position closest to the image side.

Example 7

(84) A zoom lens according to Example 7 (Numerical Example 7) illustrated in FIG. 13 includes, in order from an object side to an image side, a first lens unit U1 having a positive refractive power, a second lens unit U2 having a negative refractive power, a third lens unit U3 having a negative refractive power, a fourth lens unit U4 having a positive refractive power, and a fifth lens unit U5 having a positive refractive power. FIG. 13 illustrates a lens structure at a wide-angle end and at a state of focusing on an object at an infinite distance. The zoom lens according to Numerical Example 7 has a zoom ratio of 13.33 and satisfies the inequality (3).

(85) The first lens unit U1 is a lens unit fixed during zooming. The second to fourth lens units U2 to U4 are movable lens units each of which moves during zooming. As indicated by arrows below the second to fourth lens units U2 to U4 in the drawing, during zooming from the wide-angle end to the telephoto end, the second lens unit U2 moves to the image side, and the third lens unit U3 and fourth lens unit U4 move to the object side and then move to the image side. The fifth lens unit U5 is a final lens unit fixed during zooming and includes an aperture diaphragm SP at a position close to the object side.

(86) In the first lens unit U1, a sub unit U1f closest to the object side and a sub unit U13 third closest to the object side (closest to the image side) are fixed sub units each of which is fixed during focusing, and a sub unit U12 second closest to the object side is a movable sub unit that moves during focusing. As indicated by an arrow below the movable sub unit U12 in the drawing, the movable sub unit U12 moves to the image side during focusing from an infinite distance side to a close side.

(87) In Numerical Example 7, the positive lens U5Lp closest to the image side in the fifth lens unit U5 satisfies the inequalities (1) to (2) and the inequality (6) but does not satisfy the inequality (4). The movable lens unit having the strongest negative refractive power in the movable lens units is the second lens unit U2. The second lens unit U2 satisfies the inequality (5).

(88) The first lens unit U1 includes a negative lens U1Ln at a position closest to the object side and a positive lens U1Lp at a position closest to the image side.

(89) FIGS. 2A, 4A, 6A, 8A, 10A, 12A, and 14A illustrate longitudinal aberrations (spherical aberration, astigmatism, distortion, and chromatic aberration) of the respective zoom lenses according to Numerical Examples 1 to 7 each of which is at the wide-angle end and focuses on an object at an infinite distance. FIGS. 2B, 4B, 6B, 8B, 10B, 12B, and 14B illustrate longitudinal aberrations (spherical aberration, astigmatism, distortion, and chromatic aberration) of the respective zoom lenses according to Numerical Examples 1 to 7 each of which is at the telephoto end and focuses on an object at an infinite distance. In each spherical aberration diagram, Fno represents an F number. A solid line represents spherical aberration at a d-line (wavelength 587.6 nm), a two-dot chain line represents spherical aberration at a g-line (wavelength 435.8 nm), and a one-dot chain line represents spherical aberration at a C-line (wavelength 656.3 nm). In each astigmatism diagram, a solid line S represents a sagittal image plane, and a broken line M represents a meridional image plane. Each distortion diagram illustrates distortion at the d-line. In each chromatic aberration diagram, a two-dot chain line represents lateral chromatic aberration at the g-line, and a one-dot chain line represents lateral chromatic aberration at the C-line. ω represents a half angle of view($^{\circ}$).

(90) Hereinafter, Numerical Examples 1 to 7 are given. In each Numerical Example, a surface number i represents an order of a surface counted from the object side, r represents a curvature radius (mm) of an i-th surface counted from the object side, d represents a lens thickness or an air distance (mm) between the i-th and an (i+1)-th surface, and nd represents a refractive index at the d-line of an optical material between the i-th surface and the (i+1)-th surface. vd represents an Abbe number at the d-line of the optical material between the i-th surface and the (i+1)-th surface. BF represents a back focus (mm). A “back focus” is an air conversion length of a distance on an optical axis from a final surface (a lens surface closest to the image side) of the zoom lens to a paraxial image plane. A “total lens length” is a length acquired by adding the back focus to a distance on the optical axis from a front surface (a lens surface closest to the object side) of the zoom lens to the final surface.

(91) The Abbe number vd is expressed by the following equation where Nd, NF, and NC respectively represent refractive indexes at the d-line (587.6 nm), an F-line (486.1 nm), and the C-line (656.3 nm) of Fraunhofer lines.

$$vd = (Nd - 1) / (NF - NC)$$

(92) “*” attached to the surface number represents that the surface has an aspherical shape. The aspherical shape is expressed by the following equation where, when an optical axis direction is assumed to be an X axis, a direction orthogonal to the optical axis is assumed to be an H axis, and a light traveling direction is assumed to be positive, R represents a paraxial curvature radius, K represents a conical constant, A3, A4, A5, A6, A7, A8, A9, A10, A11, A12, A13, A14, A15, and A16 represent aspherical coefficients.

(93)

$$X = \frac{\frac{H^2}{R}}{1 + \sqrt{1 - (1 + K)(H/R)^2}} + A4H^4 + A6H^6 + A8H^8 + A10H^{10} + A12H^{12} + A14H^{14} + A16H^{16} + A3H^3 + A5H^5 + A7H^7 + A9H^9 + A11H^{11} + A13H^{13} + A15H^{15}$$

In conic constants and aspherical coefficients, “e-x” represents $\times 10.\text{sup.}-x$.

(94) Table 1 summarizes values of each inequality in Numerical Examples 1 to 7.

Numerical Example 1

(95) TABLE-US-00001 Unit: mm SURFACE DATA Surface Number r d nd vd 1 -291.688 2.80 1.80100 35.0 2 144.078 1.21 3 153.543 4.43 1.95906 17.5 4 243.217 1.68 5 168.214 12.15 1.60311 60.6 6* -244.858 10.85 7 173.088 2.50 1.83400 37.2 8 65.624 10.67 1.43875 94.7 9 340.161 6.61 10 93.619 10.70 1.43387 95.1 11 -422.601 0.20 12 84.501 9.16 1.59522 67.7 13 -5841.261 (Variable) 14 305.506 0.95 1.75500 52.3

15 18.046 7.33 16 -32.262 0.75 1.49700 81.5 17 37.702 6.51 1.83400 37.2 18 -25.260 0.43 19 -23.116 1.20 1.77250 49.6 20* -314.961 (Variable)
21 -50.584 4.31 1.80810 22.8 22 -22.449 1.10 1.80100 35.0 23 -242.799 (Variable) 24 ∞ 2.00 (Diaphragm) 25* 60.959 7.61 1.64000 60.1 26
-74.522 0.19 27 54.596 1.10 1.85478 24.8 28 34.089 4.63 1.48749 70.2 29 78.845 (Variable) 30 227.612 2.57 1.48749 70.2 31 -207.293 1.20
2.00100 29.1 32 57.206 0.98 33 53.608 5.46 1.83400 37.2 34 -137.066 43.57 35 53.272 6.12 1.43875 94.7 36 -288.236 0.96 37 40.968 11.95
1.80810 22.8 38 -36.607 0.90 1.95375 32.3 39 26.862 4.06 40 48.363 9.15 1.54814 45.8 41 -28.327 1.00 2.00100 29.1 42 78.725 -0.10 43 49.073
10.57 1.49700 81.5 44 -30.061 (Variable) Image Plane ∞ ASPHERICAL DATA 6th Surface K = -5.87800e+001 A4 = -4.11338e-007 A6 =
2.06301e-010 A8 = -8.94756e-014 A10 = 2.51858e-017 A12 = -3.21128e-021 20th Surface K = 4.38678e+002 A4 = -4.89448e-006 A6 =
-3.86929e-009 A8 = 7.15676e-011 A10 = -6.47860e-013 A12 = 2.11048e-015 25th Surface K = -5.34393e-001 A4 = -2.54409e-006 A6 =
5.8208 1e-010 A8 = 1.79765e-012 A10 = -6.91154e-015 A12 = 7.30550e-018 VARIOUS DATA 9.80 ZOOM RATIO Wide-Angle Middle
Telephoto Focal length: 25.50 78.17 250.00 F-Number: 2.79 2.80 3.70 Half Angle of View (°): 30.13 10.72 3.39 Image Height: 14.80 14.80 14.80
Total Lens Length: 316.95 316.95 316.95 BF: 37.94 37.94 37.94 d13 0.99 38.85 59.24 d20 48.23 3.58 2.77 d23 9.62 18.81 0.97 d29 10.71 8.31 6.56
d44 37.94 37.94 37.94 LENS UNIT DATA Unit Starting Surface Focal length 1 1 90.20 2 14 -22.61 3 21 -82.03 4 24 54.87 5 30 102.85

Numerical Example 2

(96) TABLE-US-00002 Unit: mm SURFACE DATA Surface Number r d nd vd 1 -175.214 3.50 1.67300 38.3 2 151.202 15.58 3 -119.783 3.61
1.43387 95.1 4 -112.613 0.20 5 227.415 13.96 1.43387 95.1 6 -129.184 9.38 7 162.176 8.02 1.43387 95.1 8 -753.576 0.08 9 125.294 8.39
1.43387 95.1 10 -915.044 0.83 11 74.473 8.42 1.43875 94.7 12* 340.725 (Variable) 13 269.294 1.30 2.00069 25.5 14 20.440 6.36 15 -39.720 1.10
1.69680 55.5 16 32.336 6.99 1.85478 24.8 17 -29.629 1.02 18 -23.763 1.10 1.72916 54.7 19* -81.231 (Variable) 20 -48.638 1.30 1.88300 40.8 21
208.585 2.06 1.95906 17.5 22 -213.735 (Variable) 23* 73.373 3.79 1.72916 54.7 24 -49.264 (Variable) 25 ∞ 6.28 (Diaphragm) 26 -44.725 3.83
1.65412 39.7 27 -19.881 1.20 1.61800 63.3 28 43.010 2.37 29 33.497 1.30 1.96300 24.1 30 22.812 6.38 1.65412 39.7 31 -58.802 15.84 32 71.434
3.35 1.95906 17.5 33 103.023 1.30 2.00100 29.1 34 28.089 1.40 35 31.130 9.86 1.43875 94.7 36 -22.965 1.40 2.00100 29.1 37 -74.760 4.89
1.49700 81.5 38 -38.132 5.45 39 -83.722 6.77 1.59270 35.3 40 -34.630 5.12 41 128.652 5.93 1.49700 81.5 42 -100.375 5.00 43 ∞ 63.04 1.60859
46.4 44 ∞ 8.70 1.51633 64.2 45 ∞ (Variable) Image Plane ∞ ASPHERICAL DATA 12th Surface K = 0.00000e+000 A4 = 8.27789e-008 A6 =
-1.49973e-011 A8 = 4.85422e-015 A10 = -5.95615e-019 19th Surface K = 1.35528e+000 A4 = -6.45504e-006 A6 = -5.00746e-009 A8 =
-1.21703e-010 A10 = 4.12728e-012 A12 = -6.25642e-014 A14 = 4.60921e-016 A16 = 1.33684e-018 23rd Surface K = 1.92930e+000 A4 =
-4.90734e-006 A6 = 5.95339e-009 A8 = -1.41239e-010 A10 = 2.19152e-012 A12 = -1.73996e-014 A14 = 6.52571e-017 A16 = -8.36889e-020
VARIOUS DATA 10.00 ZOOM RATIO Wide-Angle Middle Telephoto Focal length: 14.00 43.48 140.00 F-Number: 2.90 2.90 2.90 Half Angle 33.45
12.01 3.78 of View (°): Image Height: 9.25 9.25 9.25 Total Lens Length: 334.12 334.12 334.12 BF: 16.63 16.63 16.63 d12 1.50 38.69 58.72 d19
57.19 10.12 0.63 d22 0.99 10.68 0.97 d24 1.39 1.58 0.74 d45 16.63 16.63 16.63 LENS UNIT DATA Unit Starting Surface Focal length 1 1 77.73 2
13 -19.98 3 20 -75.96 4 23 40.78 5 25 59.39

Numerical Example 3

(97) TABLE-US-00003 Unit: mm SURFACE DATA Surface Number r d nd vd 1 -171.783 2.30 1.80100 35.0 2 360.342 3.32 3 2449.824 2.30
1.80100 35.0 4 93.547 17.69 1.49700 81.5 5 -127.730 0.40 6 164.755 8.90 1.43387 95.1 7 -323.842 6.90 8 119.088 10.53 1.61800 63.3 9
-260.350 0.15 10 65.966 5.53 1.77250 49.6 11 113.238 (Variable) 12 108.681 2.33 1.77250 49.6 13 149.799 (Variable) 14 67.372 0.90 1.88300 40.8
15 14.158 6.41 16 -64.571 7.14 1.80810 22.8 17 -13.488 0.70 1.88300 40.8 18 76.051 0.20 19 26.817 2.89 1.66680 33.0 20 63.291 (Variable) 21
-43.870 0.75 1.77250 49.6 22 30.844 3.77 1.69895 30.1 23 -80.916 2.22 24 -28.353 0.75 1.77250 49.6 25 -60.179 (Variable) 26 -122.743 3.62
1.63854 55.4 27 -33.339 0.15 28 -178.655 2.66 1.51633 64.1 29 -58.384 (Variable) 30 ∞ 1.30 (Diaphragm) 31 84.875 6.24 1.51742 52.4 32
-30.166 0.90 1.83481 42.7 33 -88.784 32.40 34 82.335 5.14 1.49700 81.5 35 -52.958 9.14 36 -451.214 1.40 1.83403 37.2 37 24.017 5.09 1.48749
70.2 38 137.787 0.36 39 64.625 7.10 1.50127 56.5 40 -22.356 1.40 1.83481 42.7 41 -69.828 3.34 42 71.892 5.20 1.48749 70.2 43 -44.146 0.50 44
-86.855 2.10 1.49700 81.5 45 -64.323 4.00 46 ∞ 33.00 1.60859 46.4 47 ∞ 13.20 1.51633 64.1 48 ∞ (Variable) Image Plane ∞ VARIOUS DATA
20.00 ZOOM RATIO Wide-Angle Middle Telephoto Focal length: 8.00 35.00 160.00 F-Number: 1.90 1.90 2.80 Half Angle of View (°): 34.51 8.76
1.97 Image Height: 5.50 5.50 5.50 Total Lens Length: 295.50 295.50 295.50 BF: 7.50 7.50 7.50 d11 0.83 26.24 37.12 d13 0.80 12.02 16.83 d20
54.59 8.56 7.32 d25 6.51 11.46 1.47 d29 0.99 5.45 0.98 d48 7.50 7.50 7.50 LENS UNIT DATA Unit Starting Surface Focal length 1 1 71.25 2 12
500.20 3 14 -14.23 4 21 -38.15 5 26 49.99 6 30 57.02

Numerical Example 4

(98) TABLE-US-00004 Unit: mm SURFACE DATA Surface Number r d nd vd 1 -161.563 2.20 1.74951 35.3 2 83.531 1.54 3 89.927 6.77
1.43387 95.1 4 183.771 0.20 5 135.868 15.24 1.43387 95.1 6 -95.350 7.88 7 136.897 6.93 1.43387 95.1 8 -454.092 0.15 9 91.750 8.76
1.43387 95.1 10 -283.987 0.15 11 53.445 4.48 1.76385 48.5 12 88.487 (Variable) 13 70.773 0.90 1.88300 40.8 14 13.760 5.02 15 -66.538 6.70
1.80810 22.8 16 -12.185 0.70 1.88300 40.8 17 52.502 0.20 18 24.812 2.78 1.62588 35.7 19 88.435 (Variable) 20 -26.441 0.70 1.83400 37.2 21
27.074 3.42 1.84649 23.9 22 -190.460 (Variable) 23 ∞ 1.96 (Diaphragm) 24 -83.419 3.00 1.63854 55.4 25 -32.030 0.15 26 11679.132 2.19 1.51823
58.9 27 -115.244 0.07 28 68.450 5.93 1.51742 52.4 29 -31.378 0.90 1.74951 35.3 30 -76.848 (Variable) 31 274.509 2.76 1.49700 81.5 32 -96.339
6.26 33 60.490 1.40 1.80100 35.0 34 22.542 5.43 1.48749 70.2 35 76.507 0.11 36 59.172 7.71 1.50127 56.5 37 -23.847 1.40 1.83481 42.7 38
-96.461 0.15 39 100.150 5.98 1.43875 94.7 40 -32.303 4.00 41 ∞ 33.00 1.60859 46.4 42 ∞ 13.20 1.51633 64.1 43 ∞ (Variable) Image Plane ∞
VARIOUS DATA 17.00 ZOOM RATIO Wide-Angle Middle Telephoto Focal length: 8.00 32.65 136.00 F-Number: 1.90 1.90 2.50 Half Angle of
View (°): 34.51 9.56 2.32 Image Height: 5.50 5.50 5.50 Total Lens Length: 271.31 271.31 271.31 BF: 7.64 7.64 7.64 d12 0.79 32.80 47.01 d19 45.91
9.49 12.81 d22 6.27 10.96 1.15 d30 40.35 40.09 32.36 d43 7.64 7.64 7.64 LENS UNIT DATA Unit Starting Surface Focal length 1 1 61.00 2 13
-13.50 3 20 -37.99 4 23 37.71 5 31 52.42

Numerical Example 5

(99) TABLE-US-00005 Unit: mm SURFACE DATA Surface Number r d nd vd 1* 78.953 2.80 1.77250 49.6 2 42.025 21.29 3 68.913 2.20
1.77250 49.6 4 44.989 20.10 5 353.478 2.20 1.77250 49.6 6 102.861 4.67 1.92286 18.9 7 143.293 10.00 8 143.296 8.54 1.49700 81.5 9*
-660.058 10.16 10 126.072 12.33 1.61800 63.3 11 -89.566 2.02 12 -385.544 2.00 1.80518 25.4 13 67.762 3.01 1.49700 81.5 14 66.178 4.38 15
93.903 10.00 1.59522 67.7 16 -116.798 0.20 17 79.466 7.23 1.73000 49.0 18 -204.448 (Variable) 19* 1243.240 1.30 1.77250 49.6 20 23.783 10.00
21 -42.047 0.90 1.77250 49.6 22 29.864 4.04 1.84666 23.8 23 -71.594 2.91 24 -18.875 0.90 1.80100 35.0 25 -26.589 (Variable) 26 -33.939 0.90
1.72916 54.7 27 54.755 2.93 1.84666 23.8 28 1294.668 (Variable) 29 ∞ 1.85 (Diaphragm) 30 700.247 4.17 1.81600 46.6 31 -56.385 0.20 32
320.448 3.94 1.51742 52.4 33 -53.596 (Variable) 34 53.543 8.45 1.48749 70.2 35 -35.854 1.20 2.00069 25.5 36 -224.787 0.20 37 28.978 3.77
1.53172 48.8 38 35.232 20.70 39 265.405 6.25 1.51823 58.9 40 -64.862 14.34 41 109.038 8.83 1.49700 81.5 42 -21.348 0.85 1.95375 32.3 43
-120.445 2.16 44 1741.197 6.19 1.92286 18.9 45 -34.707 0.85 2.00100 29.1 46 100.346 1.04 47 49.126 6.84 1.49700 81.5 48 -56.043 (Variable)
Image Plane ∞ ASPHERICAL DATA 1st Surface K = 6.16535e-001 A4 = 9.53347e-008 A6 = 2.00290e-011 A8 = -3.07226e-015 9th Surface K =
-8.09015e+002 A4 = 7.34432e-007 A6 = 1.86751e-010 A8 = -1.44173e-013 19th Surface K = -1.93517e+004 A4 = 8.47692e-006 A6 =
-7.51813e-009 A8 = 2.17579e-011 VARIOUS DATA 4.50 ZOOM RATIO Wide-Angle Middle Telephoto Focal length: 16.00 33.31 71.93 F-
Number: 2.99 3.00 3.00 Half Angle of View (°): 44.18 25.03 12.20 Image Height: 15.55 15.55 15.55 Total Lens Length: 320.01 320.01 320.01 BF:
39.96 39.96 39.96 d18 1.00 17.78 30.19 d25 29.70 7.90 2.00 d28 2.74 5.44 1.24 d33 7.77 10.08 7.77 d48 39.96 39.96 39.96 LENS UNIT DATA Unit
Starting Surface Focal length 1 1 32.00 2 19 -18.80 3 26 -50.00 4 29 37.74 5 34 95.79

Numerical Example 6

(100) TABLE-US-00006 Unit: mm SURFACE DATA Surface Number r d nd vd 1 234.095 7.49 1.43387 95.1 2 744.347 1.00 3 611.406 3.00
1.80100 35.0 4 179.646 1.07 5 180.854 14.73 1.43387 95.1 6 -616.575 11.15 7 206.490 7.85 1.43387 95.1 8 1434.650 0.20 9 143.338 5.46

1.43387 95.1 10 229.576 2.20 111 218.792 8.23 1.43387 95.1 12 293.076 (Variable) 13 157.514 1.00 1.83400 37.2 14 27.285 8.17 15 -53.185 0.90 1.77250 49.6 16 48.123 0.70 17 39.025 7.61 1.80810 22.8 18 -58.658 (Variable) 19 -36.219 1.10 1.77250 49.6 20 -16640.477 (Variable) 21 -62.809 1.30 1.71300 53.9 22 37.542 3.41 1.83400 37.2 23 123.493 (Variable) 24 622.004 3.95 1.74951 35.3 25 -65.214 1.00 26 -78.813 1.53 1.51823 58.9 27 -74.268 (Variable) 28 ∞ 1.00 (Diaphragm) 29 91.777 9.43 1.74951 35.3 30 -37.804 1.50 2.00100 29.1 31 174.214 1.00 32 84.520 8.31 1.48749 70.2 33 -39.356 1.50 1.88300 40.8 34 -60.994 50.00 35 163.915 4.64 1.51742 52.4 36 -55.320 2.54 37 36.082 1.20 1.78590 44.2 38 15.322 6.50 1.51742 52.4 39 69.483 2.01 40 100.129 5.44 1.54814 45.8 41 -2364.899 1.20 1.77250 49.6 42 29.077 0.67 43 20.807 4.04 1.49700 81.5 44 614.656 3.80 45 ∞ 34.37 1.60859 46.4 46 ∞ 13.75 1.51680 64.2 47 ∞ (Variable) Image Plane ∞ VARIOUS DATA 40.00 ZOOM RATIO Wide-Angle Middle Telephoto Focal length: 11.00 218.22 440.00 F-Number: 2.10 2.10 4.10 Half Angle of View (°): 26.57 1.44 0.72 Image Height: 5.50 5.50 5.50 Total Lens Length: 402.14 402.14 402.14 BF: 7.90 7.90 7.90 d12 10.00 108.86 113.65 d18 1.43 6.19 11.14 d20 125.30 2.96 22.17 d23 10.04 17.80 2.30 d27 3.50 14.46 1.00 d47 7.90 7.90 7.90 LENS UNIT DATA Unit Starting Surface Focal length 1 1 161.84 2 13 -60.64 3 19 -46.99 4 21 -66.60 5 24 76.81 6 28 70.45

Numerical Example 7

(101) TABLE-US-00007 Unit: mm SURFACE DATA Surface Number r d nd vd 1* 1192.402 2.50 1.83481 42.7 2 30.426 16.96 3* 137.578 2.00 1.83481 42.7 4 78.229 10.11 5 -100.411 1.80 1.83481 42.7 6 -481.398 0.15 7 93.872 3.85 1.92286 18.9 8 257.624 2.21 9 179.156 7.63 1.60300 65.4 10* -106.058 4.41 11 4935.437 8.72 1.43387 95.1 12 -55.764 0.65 13 -57.428 1.70 1.80000 29.8 14 -104.843 0.18 15 173.505 1.70 1.91650 31.6 16 56.124 12.67 1.43875 94.7 17 -148.337 0.40 18 4271.888 9.14 1.43387 95.1 19 -62.811 0.40 20 143.432 7.53 1.76385 48.5 21 -146.520 (Variable) 22 106.192 0.70 2.00100 29.1 23 18.267 4.25 24 -54.292 0.70 1.43875 94.7 25 89.426 2.07 26 -193.060 5.57 1.85478 24.8 27 -15.511 0.70 1.88300 40.8 28 140.345 0.21 29 39.231 2.87 1.64769 33.8 30 -209.632 (Variable) 31 -34.647 0.80 1.72916 54.7 32 49.836 2.41 1.84666 23.8 33 4117.439 (Variable) 34* 59.371 6.11 1.58913 61.1 35 -59.288 (Variable) 36 ∞ 2.45 (Diaphragm) 37 196.135 4.72 1.51742 52.4 38 -47.588 1.00 1.83481 42.7 39 -156.750 35.50 40 62.539 5.60 1.63980 34.5 41 -48.447 1.79 42 -70.201 0.90 1.88300 40.8 43 29.203 5.31 1.48749 70.2 44 -106.840 0.20 45 64.763 7.66 1.43875 94.7 46 -21.285 0.90 2.00100 29.1 47 -59.870 0.13 48 175.318 4.80 1.61800 63.3 49 -35.144 4.00 50 ∞ 33.00 1.60859 46.4 51 ∞ 13.20 1.51680 64.2 52 ∞ (Variable) Image Plane ∞ ASPHERICAL DATA 1st Surface K = 0.00000e+000 A4 = 4.79471e-006 A6 = 1.09101e-008 A8 = 7.85275e-012 A10 = 9.49250e-014 A12 = 1.11174e-016 A14 = 1.85317e-020 A16 = -6.47635e-026 A3 = 6.81313e-006 A5 = -1.91840e-007 A7 = -3.04740e-010 A9 = -1.17561e-012 A11 = -4.11498e-015 A13 = -1.90019e-018 A15 = -7.32675e-023 3rd Surface K = 0.00000e+000 A4 = -2.88353e-006 A6 = -7.32735e-008 A8 = -7.11623e-010 A10 = -3.23231e-013 A12 = 1.59758e-015 A14 = -6.51939e-019 A16 = -2.03836e-022 A3 = -5.37483e-006 A5 = 4.33123e-007 A7 = 9.09152e-009 A9 = 3.03607e-011 A11 = -3.27297e-014 A13 = -1.78387e-017 A15 = 2.23171e-020 10th Surface K = 0.00000e+000 A4 = 1.24811e-006 A6 = 1.37302e-008 A8 = 2.71930e-010 A10 = 2.08643e-013 A12 = -7.68000e-016 A14 = 1.05314e-018 A16 = 2.19108e-022 A3 = -1.60560e-006 A5 = -1.67398e-008 A7 = -2.65872e-009 A9 = -1.44572e-011 A11 = 1.68256e-014 A13 = -4.85868e-018 A15 = -2.64140e-020 34th Surface K = -8.55475e+000 A4 = 1.00844e-006 A6 = -2.95364e-009 A8 = 1.16661e-012 VARIOUS DATA 13.33 ZOOM RATIO Wide-Angle Middle Telephoto Focal length: 4.50 16.51 59.99 F-Number: 1.90 1.90 2.80 Half Angle of View (°): 50.71 18.43 5.24 Image Height: 5.50 5.50 5.50 Total Lens Length: 315.65 315.65 315.65 BF: 7.45 7.45 7.45 d21 0.65 37.62 53.47 d30 42.42 5.41 4.56 d33 15.45 18.86 2.11 d35 7.42 4.05 5.80 d52 7.45 7.45 7.45 LENS UNIT DATA Unit Starting Surface Focal length 1 1 30.38 2 22 -20.82 3 31 -53.01 4 34 51.14 5 36 52.07

(102) TABLE-US-00008 TABLE 1 Ineq. Example No. Inequality 1 2 3 4 5 6 7 (1) (4) vdp 81.54 81.54 81.54 94.66 81.54 81.54 63.33 (2) Ndp + 2.31 2.31 2.31 2.39 2.31 2.31 2.25 0.01vdp (3) Z 9.80 10.00 20.00 17.00 4.50 40.00 13.33 (5) |fp/fv| 1.74 5.71 34.01 4.18 2.86 0.92 2.29 (6) Le/fw 3.86 5.46 10.04 3.90 5.10 9.09 15.22

Image Pickup Apparatus

(103) FIG. 15 illustrates a configuration of a camera system (image pickup apparatus) 125 that uses, as an image pickup optical system, the zoom lens according to any one of Examples 1 to 7. A reference numeral 101 denotes the zoom lens according to any one of Examples 1 to 7. A reference numeral 124 denotes a camera, and the zoom lens 101 is an interchangeable lens that is attachable to and detachable from the camera 124.

(104) The zoom lens 101 includes a first lens unit F, a zooming unit LZ that moves during zooming, and imaging lens units R1 to R3. The first lens unit F includes a movable sub unit that moves during focusing. The zooming unit LZ includes at least three lens units each of which moves during zooming. An aperture diaphragm SP and the imaging lens units R1 to R3 are disposed on the image side of the zooming unit LZ. A lens unit IE, which is an extender or the like, is inserted between any two lens units of the imaging lens units R1 to R3 (for example, between lens units R1 and R2) from the outside of the optical path.

(105) Reference numerals 114 and 115 denote driving mechanisms including, for example, a helicoid, a cam, and the like, that respectively drive the movable sub unit in the first lens unit F and the movable lens units in the zooming unit LZ in the optical axis direction. Reference numerals 116, 117, and 118 are actuators that electrically drive the driving mechanisms 114 and 115 and the aperture diaphragm SP.

(106) Reference numerals 119, 120, and 121 denote detectors such as encoder, potentiometers, and photosensors, for detecting positions on the optical axis of the movable sub unit in the first lens unit F and the movable lens units in the zooming unit LZ and an aperture diameter of the aperture diaphragm SP.

(107) In the camera 124, a reference numeral 109 denotes a glass block corresponding to an optical filter or a color separation optical system, and a reference numeral 110 denotes an image sensor (image pickup element or photoelectric conversion element), such as a CCD sensor and a CMOS sensor, that captures (configured to pick up) an optical image (object image) of an object formed by the zoom lens 101.

(108) Reference numerals 111 and 122 denote CPUs as controllers that control the driving of the camera 124 and the zoom lens 101.

(109) When the zoom lens according to any one of Examples 1 to 7 is used as the image pickup optical system of the camera system 125, it is possible to realize high imaging performance from an optical aspect.

(110) According to the above examples, it is possible to provide, for example, a zoom lens that is beneficial to size reduction, a wide angle of view, a high zoom ratio, a large diameter ratio, and high optical performance.

(111) While the disclosure has been described with reference to exemplary embodiments, it is to be understood that the disclosure is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

(112) This application claims the benefit of Japanese Patent Application No. 2021-131582, filed on Aug. 12, 2021, which is hereby incorporated by reference herein in its entirety.

Claims

1. A zoom lens comprising, in order from an object side to an image side: a first lens unit having a positive refractive power and configured not to move for zooming; at least three movable lens units configured to move so that a distance between each pair of adjacent lens units therein changes for zooming; and a final lens unit disposed closest to the image side in lens units included in the zoom lens, having a positive refractive power, and configured not to move for zooming, wherein a movable lens unit closest to the image side in the at least three movable lens units has a positive refractive power, and a movable lens unit second closest to the image side in the at least three movable lens units has a negative refractive power, wherein the first lens unit includes: a first sub unit closest to the object side and configured not to move for focusing; and a second sub unit configured to move for focusing, wherein the final lens unit includes a positive lens closest to the image side in elements having a refractive power in the final lens unit, and wherein following inequalities are satisfied:

$80 \leq vdp \leq 81.54$,

$2.24 \leq Ndp + 0.01vdp$,

$4 \leq Z \leq 50$, and

$2.5 \leq Le/fw \leq 5.46$ where vdp represents an Abbe number with respect to a d-line of the positive lens, Ndp represents a refractive index with respect to the d-line of the positive lens, Z represents a zoom ratio of the zoom lens, Le represents a distance on an optical axis from a lens surface closest to the object side in the final lens unit to an image side surface of the positive lens in the final lens unit, and fw represents a focal length of the zoom lens at a wide-angle end.

2. The zoom lens according to claim 1, wherein the final lens unit includes four or more lenses each of which has a refractive power.

3. The zoom lens according to claim 1, wherein a following inequality is satisfied:

$0.5 \leq |fp/fv| \leq 50.0$ where fp represents a focal length of the positive lens, and fv represents a focal length of a movable lens unit having a strongest negative refractive power in the at least three movable lens units.

4. The zoom lens according to claim 1, wherein a movable lens unit having a strongest negative refractive power in the at least three movable lens units includes two negative lenses and one positive lens.

5. The zoom lens according to claim 1, wherein the first lens unit includes a positive lens at a position closest to the image side in the first lens unit.

6. The zoom lens according to claim 1, wherein a movable lens unit having a strongest negative refractive power in the at least three movable lens units moves to the image side for zooming from a wide-angle end to a telephoto end.

7. The zoom lens according to claim 1, wherein the zoom lens includes, as the at least three movable lens units, in order from the object side to the image side: a second lens unit having a negative refractive power; a third lens unit having a negative refractive power; and a fourth lens unit having a positive refractive power, and wherein the zoom lens includes, as the final lens unit, a fifth lens unit having a positive refractive power.

8. The zoom lens according to claim 1, wherein the zoom lens includes, as the at least three movable lens units, in order from the object side to the image side: a second lens unit having a positive or negative refractive power; a third lens unit having a negative refractive power; a fourth lens unit having a negative refractive power; and a fifth lens unit having a positive refractive power, wherein the zoom lens includes, as the final lens unit, a sixth lens unit having a positive refractive power.

9. An apparatus comprising: a zoom lens; and a pickup element configured to pick up an image formed by the zoom lens, wherein the zoom lens includes, in order from an object side to an image side: a first lens unit having a positive refractive power and configured not to move for zooming; at least three movable lens units configured to move so that a distance between each pair of adjacent lens units therein changes for zooming; and a final lens unit disposed closest to the image side in lens units included in the zoom lens, having a positive refractive power, and configured not to move for zooming, wherein a movable lens unit closest to the image side in the at least three movable lens units has a positive refractive power, and a movable lens unit second closest to the image side in the at least three movable lens units has a negative refractive power, wherein the first lens unit includes: a first sub unit closest to the object side and configured not to move for focusing; and a second sub unit configured to move for focusing, wherein the final lens unit includes a positive lens closest to the image side in elements having a refractive power in the final lens unit, wherein following inequalities are satisfied:

$80 \leq vdp \leq 81.54$,

$2.24 \leq Ndp + 0.01vdp$,

$4 \leq Z \leq 50$, and

$2.5 \leq Le/fw \leq 5.46$ where vdp represents an Abbe number with respect to a d-line of the positive lens, Ndp represents a refractive index with respect to the d-line of the positive lens, Z represents a zoom ratio of the zoom lens, Le represents a distance on an optical axis from a lens surface closest to the object side in the final lens unit to an image side surface of the positive lens in the final lens unit, and fw represents a focal length of the zoom lens at a wide-angle end.

10. The apparatus according to claim 9, wherein the final lens unit includes four or more lenses each of which has a refractive power.

11. The apparatus according to claim 9, wherein a following inequality is satisfied:

$0.5 \leq |fp/fv| \leq 50.0$ where fp represents a focal length of the positive lens, and fv represents a focal length of a movable lens unit having a strongest negative refractive power in the at least three movable lens units.

12. The apparatus according to claim 9, wherein a movable lens unit having a strongest negative refractive power in the at least three movable lens units includes two negative lenses and one positive lens.

13. The apparatus according to claim 9, wherein the first lens unit includes a positive lens at a position closest to the image side in the first lens unit.

14. The apparatus according to claim 9, wherein a movable lens unit having a strongest negative refractive power in the at least three movable lens units moves to the image side for zooming from a wide-angle end to a telephoto end.

15. The apparatus according to claim 9, wherein the zoom lens includes, as the at least three movable lens units, in order from the object side to the image side: a second lens unit having a negative refractive power; a third lens unit having a negative refractive power; and a fourth lens unit having a positive refractive power, and wherein the zoom lens includes, as the final lens unit, a fifth lens unit having a positive refractive power.

16. The apparatus according to claim 9, wherein the zoom lens includes, as the at least three movable lens units, in order from the object side to the image side: a second lens unit having a positive or negative refractive power; a third lens unit having a negative refractive power; a fourth lens unit having a negative refractive power; and a fifth lens unit having a positive refractive power, wherein the zoom lens includes, as the final lens unit, a sixth lens unit having a positive refractive power.

17. A zoom lens comprising, in order from an object side to an image side: a first lens unit having a positive refractive power and configured not to move for zooming; at least three movable lens units configured to move so that a distance between each pair of adjacent lens units therein changes for zooming; and a final lens unit disposed closest to the image side in lens units included in the zoom lens, having a positive refractive power, and configured not to move for zooming, wherein a movable lens unit closest to the image side in the at least three movable lens units has a positive refractive power, and a movable lens unit second closest to the image side in the at least three movable lens units has a negative refractive power, wherein a movable lens unit having a strongest negative refractive power in the at least three movable lens units includes two negative lenses and one positive lens, wherein the first lens unit includes: a first sub unit closest to the object side and configured not to move for focusing; and a second sub unit configured to move for focusing, wherein the final lens unit includes a positive lens closest to the image side in elements having a refractive power in the final lens unit, and wherein following inequalities are satisfied:

$62 \leq vdp \leq 81.54$,

$2.24 \leq Ndp + 0.01vdp$,

$4 \leq Z \leq 50$, and

$2.5 \leq Le/fw \leq 5.46$ where vdp represents an Abbe number with respect to a d-line of the positive lens, Ndp represents a refractive index with respect to the d-line of the positive lens, Z represents a zoom ratio of the zoom lens, Le represents a distance on an optical axis from a lens surface closest to the object side in the final lens unit to an image side surface of the positive lens in the final lens unit, and fw represents a focal length of the zoom lens at a wide-angle end.

18. An apparatus comprising: a zoom lens; and a pickup element configured to pick up an image formed by the zoom lens, wherein the zoom lens includes, in order from an object side to an image side: a first lens unit having a positive refractive power and configured not to move for zooming; at least three movable lens units configured to move so that a distance between each pair of adjacent lens units therein changes for zooming; and a final lens unit disposed closest to the image side in lens units included in the zoom lens, having a positive refractive power, and configured not to move for

zooming, wherein a movable lens unit closest to the image side in the at least three movable lens units has a positive refractive power, and a movable lens unit second closest to the image side in the at least three movable lens units has a negative refractive power, wherein a movable lens unit having a strongest negative refractive power in the at least three movable lens units includes two negative lenses and one positive lens, wherein the first lens unit includes: a first sub unit closest to the object side and configured not to move for focusing; and a second sub unit configured to move for focusing, wherein the final lens unit includes a positive lens closest to the image side in elements having a refractive power in the final lens unit, wherein following inequalities are satisfied:

$$62 \leq vdp \leq 81.54,$$

$$2.24 \leq Ndp + 0.01vdp,$$

$4 \leq Z \leq 50$, and $2.5 \leq Le/fw \leq 5.46$ where vdp represents an Abbe number with respect to a d-line of the positive lens, Ndp represents a refractive index with respect to the d-line of the positive lens, Z represents a zoom ratio of the zoom lens, Le represents a distance on an optical axis from a lens surface closest to the object side in the final lens unit to an image side surface of the positive lens in the final lens unit, and fw represents a focal length of the zoom lens at a wide-angle end.
